Ch 26: Current and Resistance

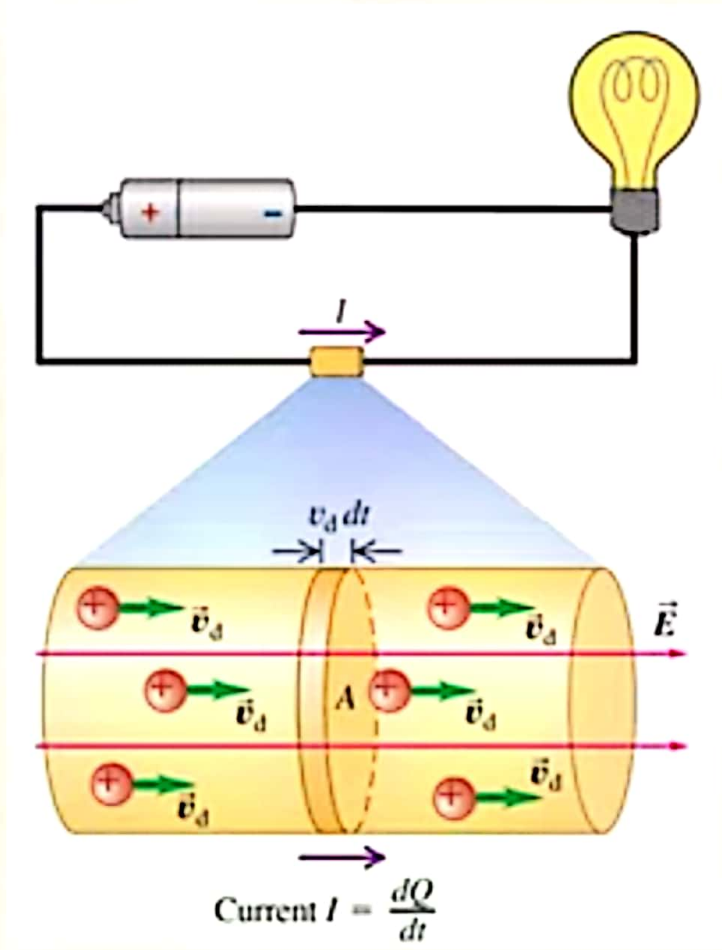
Current: Electric current in a wire is defined as the amount of charge passing through any cross section of the wire per unit time:

$$I = \frac{\Delta Q}{\Delta t}$$

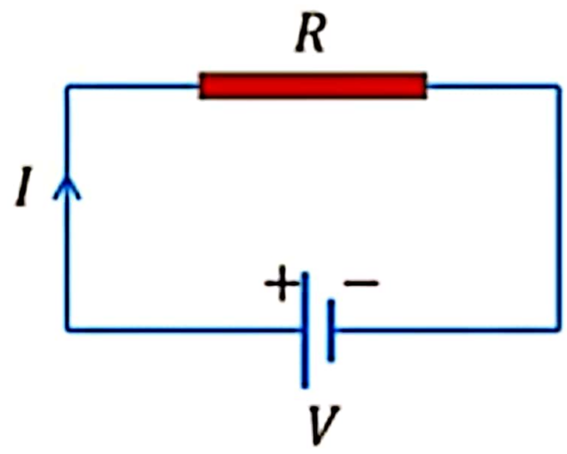
Or more precisely, the instantaneous current is:

$$I = \frac{dQ}{dt}$$

A **circuit** made of a battery and a light bulb.



Current, Voltage, and Resistance, The Ohm's Law



In the circuit shown a sample of a material is connected to a battery of voltage V. Experiments show that a current I passes through the circuit that is proportional to the voltage.

$$V \propto I$$

The proportionality factor is defined to be the resistance of the sample denoted by R.

$$V = RI$$

The unit of R is Volt/Amp, which is called an **Ohm**, denoted by the Greek capital letter **Ω** (omega)

$$1\,\Omega = 1\,\frac{\text{Volt}}{\text{Amp}} = 1\,\frac{\text{V}}{\text{A}}$$

The formula $V = RI$ is called the **Ohm's Law**.

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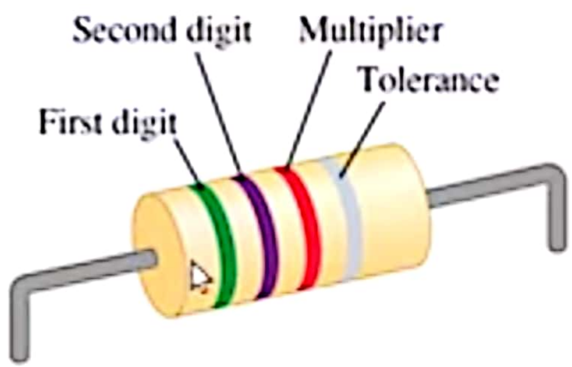


Color Codes for Resistors

Color	Value as Digit	Value as Multiplier
Black	0	1
Brown	1	10
Red	2	10^2
Orange	3	10^3
Yellow	4	10^4
Green	5	10^5
Blue	6	10^6
Violet	7	10^7
Gray	8	10^8
White	9	10^9

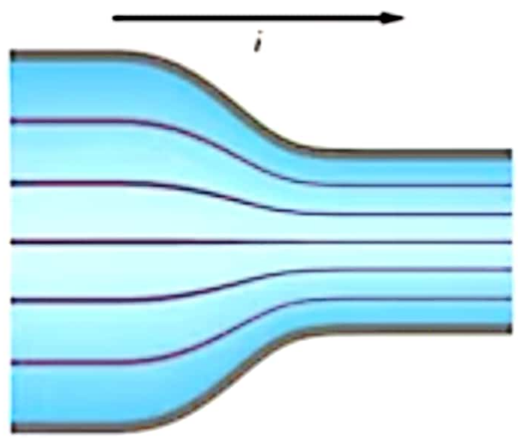
Current Density

The current density of a uniform current I through a wire of cross-sectional area A is denoted by J, and is defined by the current through the cross section divided by the cross-sectional area

$$J = \frac{I}{A}$$

The units of J are Amps per square-meters

$$\text{Units of } J = \frac{\text{A}}{\text{m}^2}$$



In general, The current density is a vector in the direction of the current at any point. The net current through any surface, S is given by:

$$I = \int_S \vec{\mathbf{J}} \cdot d\vec{\mathbf{A}}$$

Resistivity and Microscopic Ohm's Law

The electric potential across a sample sets up an electric field along the length of the sample. This electric field causes the electrons to move (in the opposite direction) and as a result a current density J is created at any point inside the material which is proportional to the electric field intensity E.

$$E \propto J$$

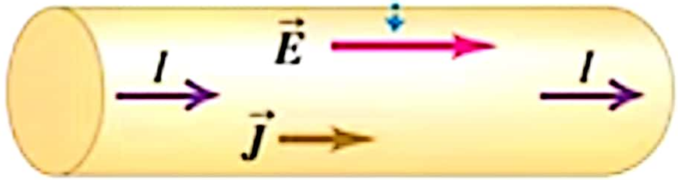
The proportionality factor is defined to be the **resistivity** of the material denoted by ρ. In Vector form we have

$$\boxed{\vec{\mathbf{E}} = \rho\,\vec{\mathbf{J}}}$$

This is called the **Microscopic** version of the **Ohm's Law**. The inverse of resistivity ρ is called the conductivity of the material denoted by σ. So $\sigma = 1/\rho$. The units of resistivity are:

$$\text{Units of Resistivity } = \Omega\,.\,\text{m}$$

An electric field E produced inside a conductor causes a current.



Unit of $\rho = \dfrac{V/m}{A/m^2} = \dfrac{V}{A} m$

$= \Omega \cdot m$

Resistivity and Microscopic Ohm's Law

Resistivity ρ. $\boxed{\vec{\mathbf{E}} = \rho\,\vec{\mathbf{J}}}$

	Substance	ρ $(\Omega \cdot m)$
Conductors		
Metals	Silver	1.47×10^{-8}
	Copper	1.72×10^{-8}
	Gold	2.44×10^{-8}
	Aluminum	2.75×10^{-8}
	Tungsten	5.25×10^{-8}
	Steel	20×10^{-8}
	Lead	22×10^{-8}
	Mercury	95×10^{-8}
Alloys	Manganin (Cu 84%, Mn 12%, Ni	44×10^{-8}
	Constantan (Cu 60%, Ni 40%)	49×10^{-8}
	Nichrome	100×10^{-8}

	Substance	ρ $(\Omega \cdot m)$
Semiconductors		
	Pure carbon (graphite)	3.5×10^{-5}
	Pure germanium	0.60
	Pure silicon	2300
Insulators		
	Amber	5×10^{14}
	Glass	10^{10}–10^{14}
	Lucite	$>10^{13}$
	Mica	10^{11}–10^{15}
	Quartz (fused)	75×10^{16}
	Sulfur	10^{15}
	Teflon	$>10^{13}$
	Wood	10^{8}–10^{11}

Resistivity and Resistance

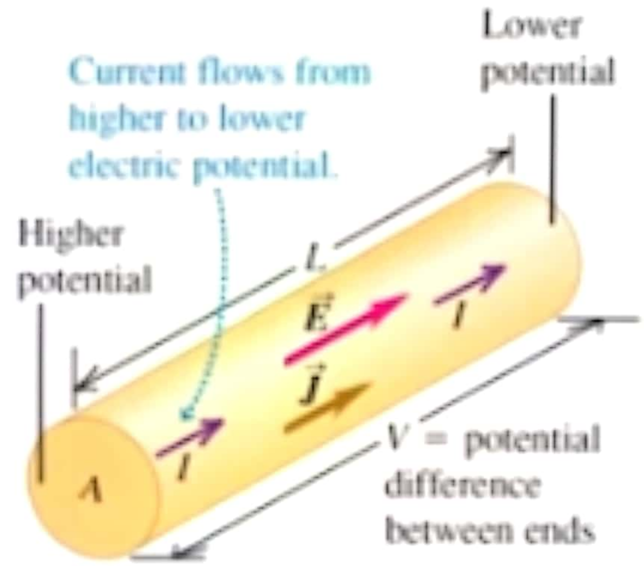
Consider a cylinder of length L, cross-sectional area A, made of a material of resistivity ρ is subjected to a potential difference V across its length. From the microscopic Ohm's Law we have $E = \rho J$. But $E = V/L$, and the current density is $J = I/A$. So

$$\frac{V}{L} = \frac{\rho I}{A}$$

Replacing V by RI and solving for R we have:

$$\boxed{R = \rho \frac{L}{A}}$$

This useful formula gives the resistance of a cylindrical sample in terms of its geometry (length and cross-sectional area) and the resistivity of its material.



Drift Velocity

The current in a wire is a result of the average velocity of the charges in the wire that is referred to as the **drift velocity**. It depends on the magnitude of the current density J at each point, the charge q of each charge carrier and the number of charge carriers per unit volume n.

Consider the short section of the wire shown in the figure. The number of charge carriers in the section of length $dx = v_d dt$ is

$$N = nAv_d dt$$

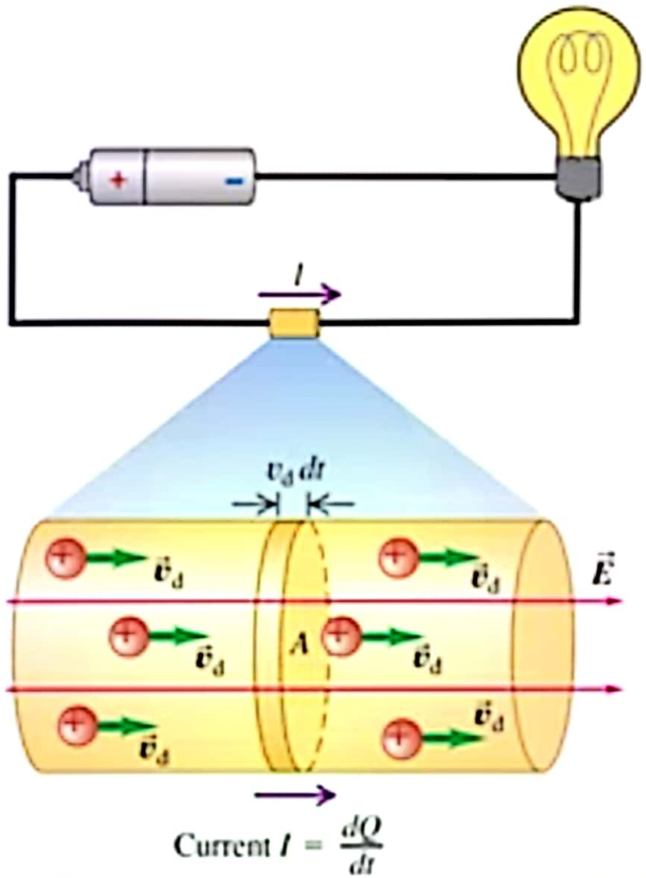
And the amount of moving charge in that section is $dQ = Nq$:

$$dQ = nqAv_d dt$$

And the current is $I = dQ/dt$:

$$\boxed{I = nqAv_d}$$

And the current density $J = I/A$: $\boxed{J = nqv_d}$



Drift Velocity

Example: Find the drift speed of electrons in a copper wire of diameter $D = 1.0$ mm carrying a current of 1.0 Amp.

To find the drift speed we need to find the number of charged carriers per unit volume for copper. We use the atomic mass of copper = 63.5 g/mol, and the mass density of copper =8960 kg/m^3. Assuming that each copper atom contributes one conduction electron to the sample the number of charge carriers is the same as the number of atoms in unit volume.

$$n = \frac{8960\ \mathrm{kg/m^3}}{0.0635\ \mathrm{kg/mol}} \times 6.02 \times 10^{23}\ \mathrm{atoms/mol}$$

$$n = 8.49 \times 10^{28}\ \mathrm{atoms/m^3}$$

$$\boxed{I = nqAv_d}$$

$$v_d = \frac{I}{nqA} = \frac{1}{(8.49 \times 10^{28})(1.6 \times 10^{-19})\left(\frac{\pi}{4}\right)(10^{-3})^2}$$

$$v_d = 0.094\ \mathrm{mm/s}$$